Status	Finished
Started	Friday, 16 October 2026, 11:11 AM
Completed	Friday, 16 October 2026, 11:16 AM
Duration	4 mins 58 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. 1
c. n ✓
d. n/2

Question **2**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all
b. When the item is not the array at all
c. When the item is somewhere in the middle of the array ✓
d. When the item is the last element in the array

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
b. Must use a sorted array
c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
d. There must be a mechanism to access middle element directly

Question **4**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100
b. 94 and 99
c. 89 and 94
d. 90 and 99 ✓

Question **5**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 10 ✓
c. 1000
d. 500

Question **6**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. The search is considered unsuccessful ✓
c. An error is raised
d. The last element is returned

Question **7**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Binary search ✓
c. Hash search
d. Sequential search

Question **8**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Bubble ✓

c. Complexity

d. Insertion

Question **9**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the search algorithm finishes

b. When the list is sorted

c. When the element is found in the list ✓

d. When the list contains duplicate elements

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. At the beginning of the list

b. In the entire list

c. In the right sublist

d. In the left sublist ✓

Question **11**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding
b. Discovering
c. Mining
d. Searching ✓

Question **12**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts
b. Compare the target element with the first element in the list
c. Compare the target element with the middle element in the list
d. Sort the list ✗

Question **13**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n
b. n ✓
c. n/2
d. n-1

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓
b. When the list is unsorted
c. When the list is very small
d. When the list contains duplicate elements

Question **15**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It can be applied only if the list is sorted ✓
- b. It always starts from the beginning of the list
- c. It compares elements sequentially
- d. It works on unsorted lists